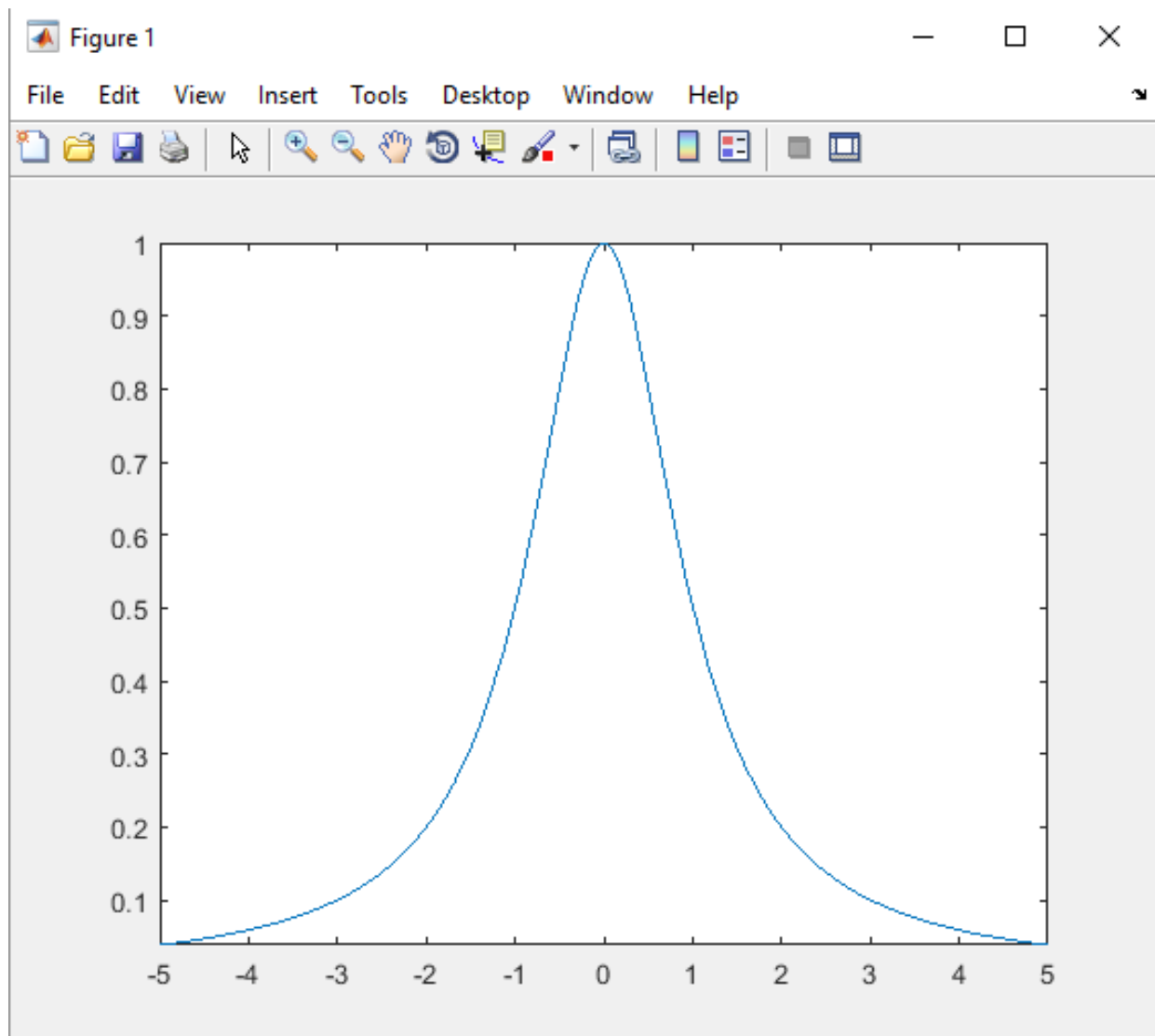


# Computer Homework 6.2

## Problem 1

For  $n = 5, 10$  and  $15$ , find the Newton interpolating polynomial  $p_n$  for the function  $f(x) = 1/(1 + x^2)$  on the interval  $[-5, 5]$ . Use equally spaced nodes. In each case, compute  $f(x) - p_n(x)$  for 30 equally spaced points in  $[-5, 5]$  to see the divergence of  $p_n$  from  $f$ .

$$f(x) = \frac{1}{1 + x^2}, x \in [-5, 5]$$



Using the following program:

```
function Pnx = Pn(F,L,U,N) % This function will take in a function F,  
% an Lower Bound L, an Upper Bound U, and a number of nodes N and will use  
% Newton Divided difference to generate a polynomial.  
% Then this function will plot the polynomial of degree of most N and an  
% array of the error between the function and the polynomial at 30 points.  
  
delta = (U-L)/N; %This will be the step size between each nodes.  
Coeff = []; % This array will contain the coefficients.  
X = []; % This array will be used to hold the node points.  
  
for i=0:N % Filling up the X array with our node points.  
    AX = [L + i*delta];  
    X = [X AX];  
end  
  
% Using the algorithim found on page 332.  
for i=0:N  
    Coeff = [Coeff F(X(i+1))]; %Starts with the functional values.  
end  
  
% This loop will iterate through the functional values and alter the values  
% inside the array in a recursive way so that it will arrive at the each  
% divided difference value as the array increases in index.  
for j = 1:N  
    n = N;  
    while (n>=j)  
        k = n+1;  
        Coeff(k) = (Coeff(k)-Coeff(k-1))/(X(k)-X(k-j));  
        n = n-1;  
    end  
end  
  
% Here we will calculate some values so that we can plot the graph and so  
% that we can see the error between the polynomial and the original  
% function at 30 points.  
% The reason for using 290 is because since we want 30, points, we will  
% need 29 equally spaced intervals and since we want to see the polynomial  
% well, we will have a high number of points to see what the function is  
% doing.  
l = L; % Starting point of our graph.  
o = 1;  
XX = [l]; % Starting point.  
FX = [F(l)]; % Starting value.  
delta290 = (U-L)/290; % Breaking up the domain into small intervals.  
Error = [0]; % At the initial point, the error is 0 (Node point).  
for v = 1:290 % Going through the domain.  
    l = l + delta290;  
    XX = [XX l]; % Add a point in its domain.  
  
    fx = Coeff(1); % Starting Pn with its constant term.  
  
    % This represents the increasing product of differences between the  
    % point x and the nodes xi.
```

```
for i = 1:(length(Coeff)-1)
    P = 1;
    for j = 1:i
        P = P*(1-X(j));
    end
    % Adding the prod(Ci(x-xI)) to C0 and lower terms.
    fx = fx + Coeff(i + 1)*P;
end

% Since we wanted to know how the polynomial diverges from the original
% function, we check the difference at every 10 points.
if (mod(v,10)) == 0
    Error = [Error abs(F(1)-fx)];
end
% Adding the function values to the range array.
FX = [FX fx];
end

plot(XX,FX); % Plots the domain and range array to create a graph of Pn.
%This array contains the errors at the 30 evenly spaced points.
Pnx = Error;
```

## Results:

```
>> f = @(x) 1/(1+x^2);
```

```
>> Pn(f,-5,5,5)
```

Error =

Columns 1 through 8

|   |        |        |        |        |        |        |        |
|---|--------|--------|--------|--------|--------|--------|--------|
| 0 | 0.0740 | 0.1062 | 0.1056 | 0.0806 | 0.0394 | 0.0102 | 0.0602 |
|---|--------|--------|--------|--------|--------|--------|--------|

Columns 9 through 16

|        |        |        |        |        |        |        |        |
|--------|--------|--------|--------|--------|--------|--------|--------|
| 0.1020 | 0.1256 | 0.1183 | 0.0635 | 0.0567 | 0.2400 | 0.4059 | 0.4059 |
|--------|--------|--------|--------|--------|--------|--------|--------|

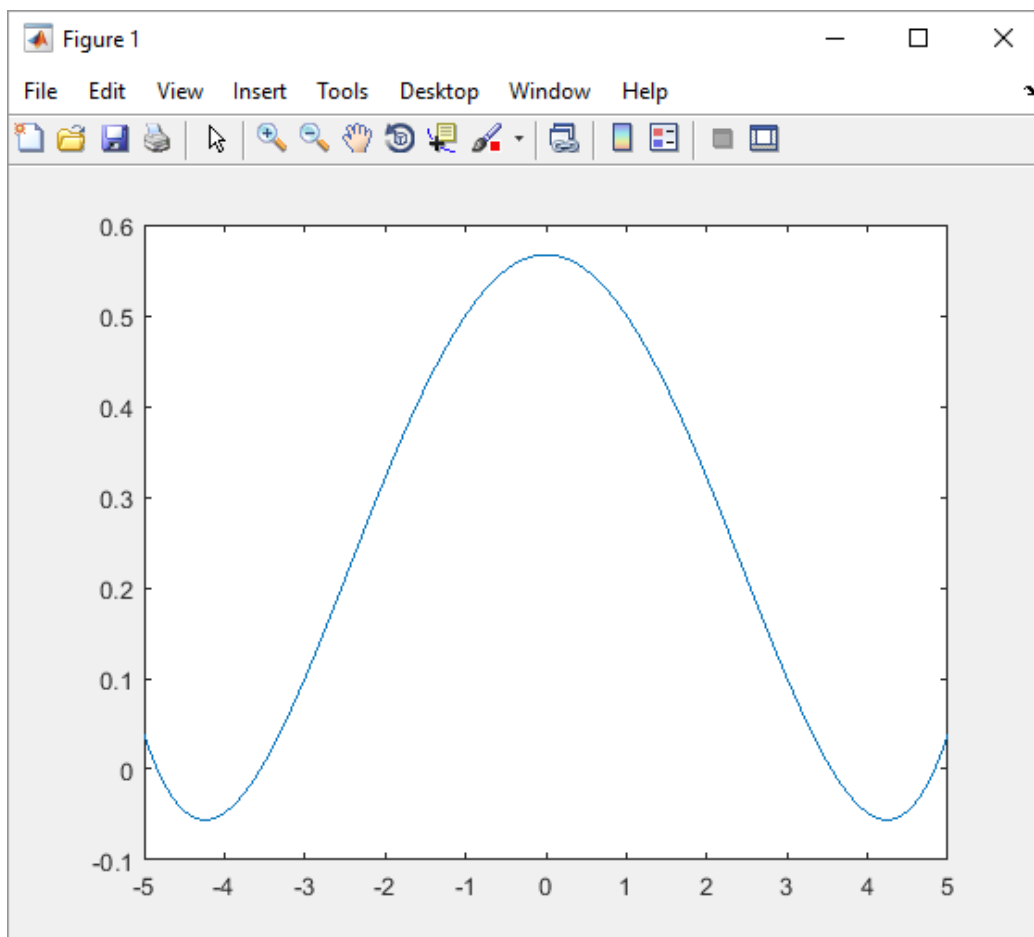
Columns 17 through 24

|        |        |        |        |        |        |        |        |
|--------|--------|--------|--------|--------|--------|--------|--------|
| 0.2400 | 0.0567 | 0.0635 | 0.1183 | 0.1256 | 0.1020 | 0.0602 | 0.0102 |
|--------|--------|--------|--------|--------|--------|--------|--------|

Columns 25 through 30

|        |        |        |        |        |        |
|--------|--------|--------|--------|--------|--------|
| 0.0394 | 0.0806 | 0.1056 | 0.1062 | 0.0740 | 0.0000 |
|--------|--------|--------|--------|--------|--------|

$$P_5(x)$$



```
>> Pn(f,-5,5,10)
```

Error =

Columns 1 through 8

|   |        |        |        |        |        |        |        |
|---|--------|--------|--------|--------|--------|--------|--------|
| 0 | 1.8897 | 0.8728 | 0.0629 | 0.3275 | 0.1764 | 0.0351 | 0.1186 |
|---|--------|--------|--------|--------|--------|--------|--------|

Columns 9 through 16

|        |        |        |        |        |        |        |        |
|--------|--------|--------|--------|--------|--------|--------|--------|
| 0.0678 | 0.0266 | 0.0725 | 0.0402 | 0.0247 | 0.0444 | 0.0090 | 0.0090 |
|--------|--------|--------|--------|--------|--------|--------|--------|

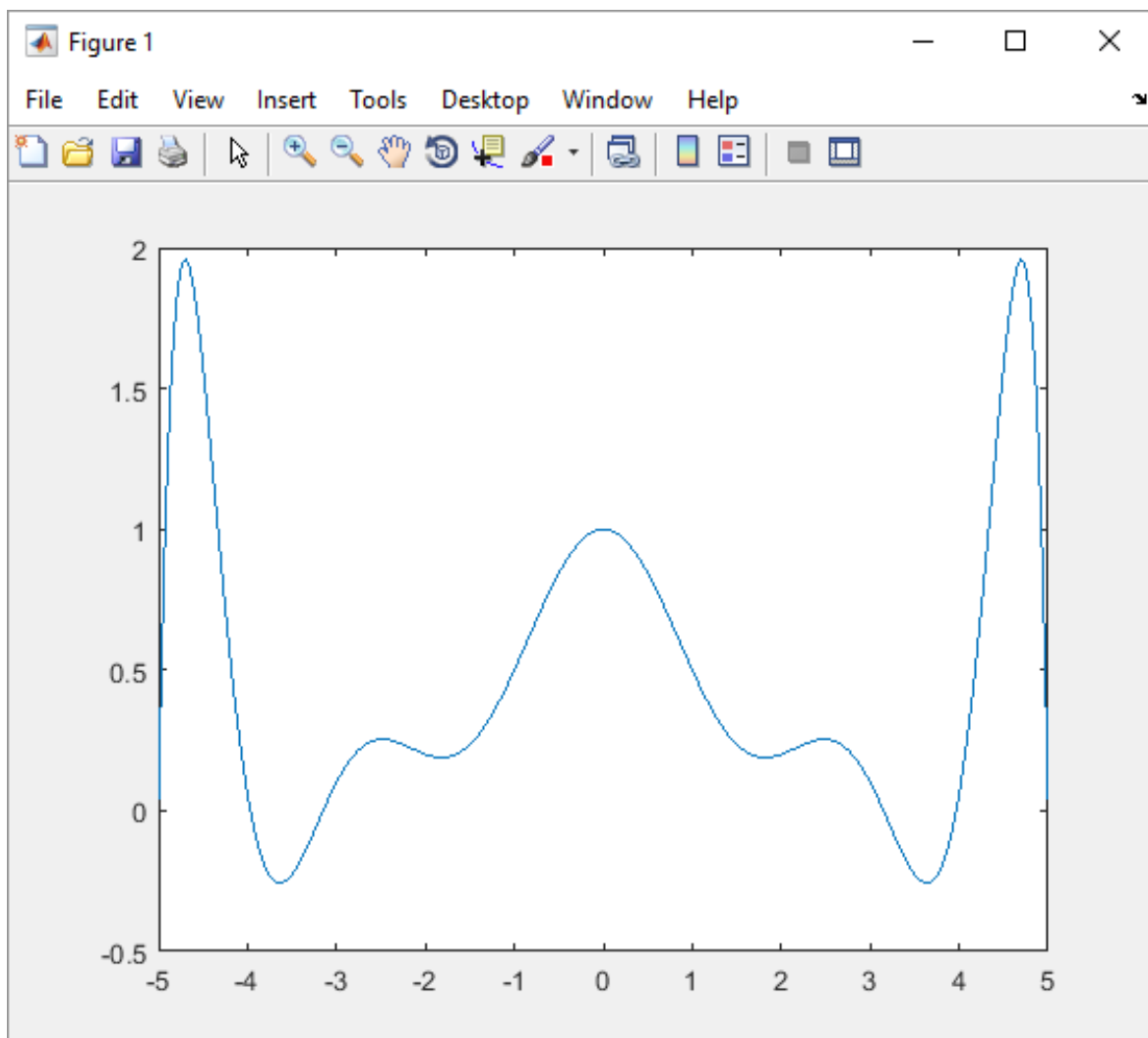
Columns 17 through 24

|        |        |        |        |        |        |        |        |
|--------|--------|--------|--------|--------|--------|--------|--------|
| 0.0444 | 0.0247 | 0.0402 | 0.0725 | 0.0266 | 0.0678 | 0.1186 | 0.0351 |
|--------|--------|--------|--------|--------|--------|--------|--------|

Columns 25 through 30

|        |        |        |        |        |        |
|--------|--------|--------|--------|--------|--------|
| 0.1764 | 0.3275 | 0.0629 | 0.8728 | 1.8897 | 0.0000 |
|--------|--------|--------|--------|--------|--------|

$P_{10}(x)$



```
>> Pn(f,-5,5,15)
```

Error =

Columns 1 through 8

|   |        |        |        |        |        |        |        |
|---|--------|--------|--------|--------|--------|--------|--------|
| 0 | 1.5055 | 0.0537 | 0.1978 | 0.0204 | 0.0497 | 0.0101 | 0.0204 |
|---|--------|--------|--------|--------|--------|--------|--------|

Columns 9 through 16

|        |        |        |        |        |        |        |        |
|--------|--------|--------|--------|--------|--------|--------|--------|
| 0.0071 | 0.0128 | 0.0071 | 0.0121 | 0.0102 | 0.0154 | 0.0160 | 0.0160 |
|--------|--------|--------|--------|--------|--------|--------|--------|

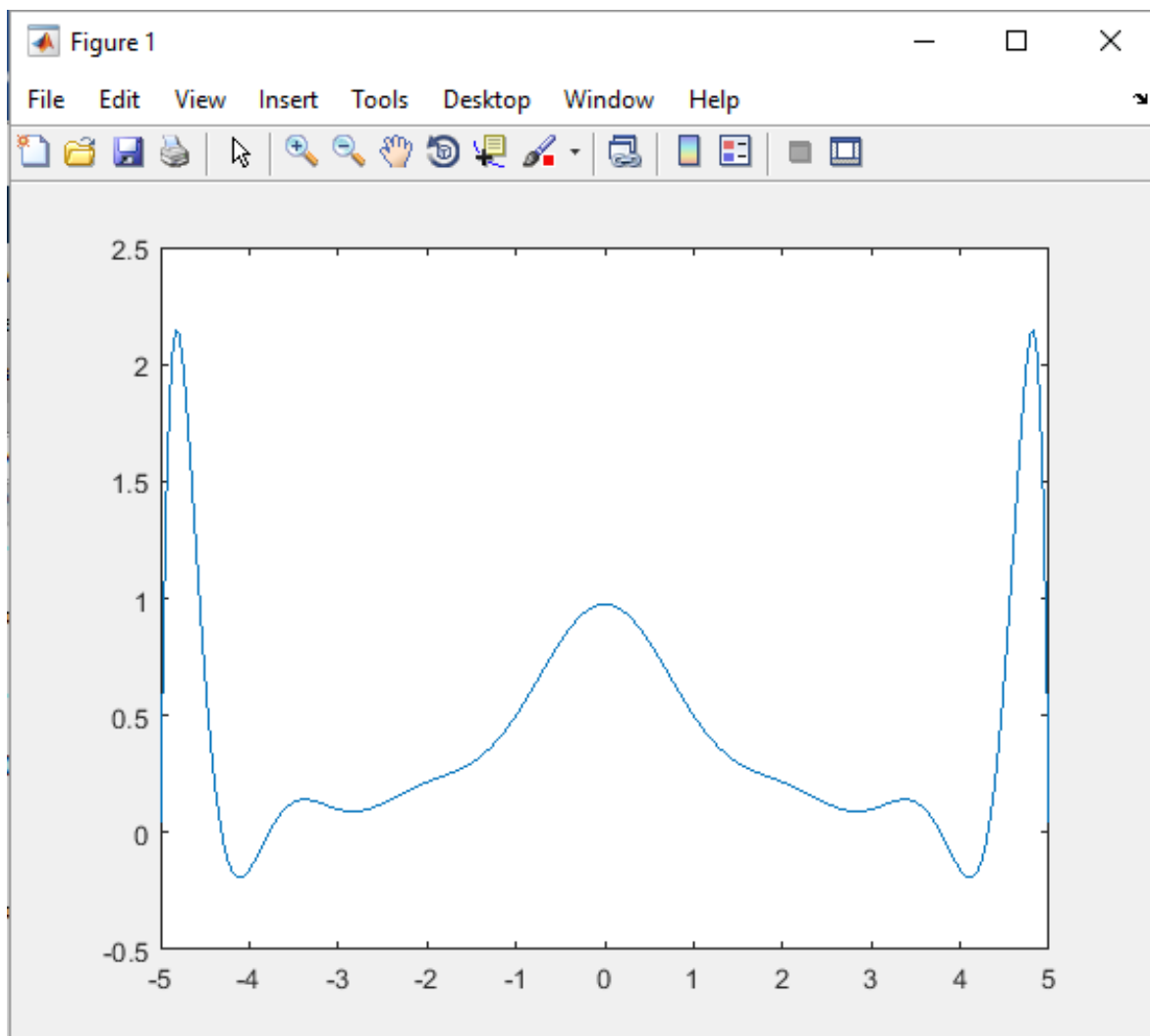
Columns 17 through 24

|        |        |        |        |        |        |        |        |
|--------|--------|--------|--------|--------|--------|--------|--------|
| 0.0154 | 0.0102 | 0.0121 | 0.0071 | 0.0128 | 0.0071 | 0.0204 | 0.0101 |
|--------|--------|--------|--------|--------|--------|--------|--------|

Columns 25 through 30

|        |        |        |        |        |        |
|--------|--------|--------|--------|--------|--------|
| 0.0497 | 0.0204 | 0.1978 | 0.0537 | 1.5055 | 0.0000 |
|--------|--------|--------|--------|--------|--------|

$P_{15}(x)$



Suppressing the plot and doing the following:

```
>> E5 = Pn(f,-5,5,5);  
>> E10 = Pn(f,-5,5,10);  
>> E15 = Pn(f,-5,5,15);  
>> norm(E5)  
ans = 0.7867  
>> norm(E10)  
ans = 3.0017  
>> norm(E15)  
ans = 2.1507
```

Note that it seems that the norm improved at  $n = 15$ , so we would be mistaken to think it's getting better, but:

```
>> E20 = Pn(f,-5,5,20);  
>> norm(E20)  
ans = 28.2739
```

Thus, we can see that we are diverging from the function on points not nodal. The reason for the improvement in  $P_{15}$  is because the points tested were close to the node points thus they aren't going to be too divergent. Looking at the graph will give us a better idea of what the divergence really is, and we saw that it was worse for  $P_{15}$  than  $P_{10}$ .